

Research Protocol v1.2

Digital Transformation and Its Impact on the Quality and Efficiency of Higher Education: A Case Study of the Universidad Nacional Mayor de San Marcos

1. Project Title

Research Focus

This project investigates the influence of digital transformation on educational quality and institutional efficiency within the Universidad Nacional Mayor de San Marcos (UNMSM). The study examines digital transformation as a strategic organizational capability that extends beyond technology adoption, encompassing governance, academic innovation, digital culture, institutional leadership, and data-driven decision-making. By adopting a mixed-methods case study design, the research seeks to generate empirical evidence that informs institutional modernization and contributes to the scientific literature on higher education transformation in emerging economies.

2. Problem Statement

Digital transformation has become one of the defining challenges for higher education institutions worldwide. Universities are expected to modernize academic, administrative, and governance processes while simultaneously improving educational quality, institutional efficiency, and stakeholder satisfaction. Although significant investments have been made in digital infrastructure, many institutions continue to struggle with integrating technological innovation into their organizational culture and strategic decision-making.

In Peru, public universities are undergoing digital modernization driven by national policies, accreditation requirements, and increasing societal expectations. Nevertheless, empirical evidence explaining how digital transformation contributes to measurable improvements in educational quality and institutional efficiency remains limited. Most existing studies emphasize technology adoption rather than organizational outcomes, leaving an important research gap.

The Universidad Nacional Mayor de San Marcos represents an ideal institutional context for investigating these relationships because of its strategic role within the Peruvian higher education system and its ongoing digital transformation initiatives. Consequently, understanding how digital

transformation influences institutional performance is both academically significant and practically relevant.

3. Research Questions

Main Research Question

How does digital transformation influence the quality and efficiency of higher education at the Universidad Nacional Mayor de San Marcos?

Supporting Research Questions

1. Which dimensions of digital transformation have the greatest influence on educational quality?
2. To what extent does digital transformation improve institutional efficiency?
3. Which organizational and technological factors facilitate or hinder successful digital transformation?
4. Which evidence-based strategies can strengthen digital transformation in public universities?

4. Research Objectives

General Objective

To evaluate the influence of digital transformation on educational quality and institutional efficiency at the Universidad Nacional Mayor de San Marcos.

Specific Objectives

- Assess the maturity of digital transformation within the institution.
- Examine the relationship between digital transformation and educational quality.
- Analyze the influence of digital transformation on institutional efficiency.
- Identify organizational enablers and barriers affecting digital transformation.
- Develop strategic recommendations for sustainable institutional digital transformation.

5. Literature Review Summary

The literature conceptualizes digital transformation as a multidimensional organizational phenomenon integrating technological innovation, organizational change, leadership, governance, digital competencies, and institutional resilience.

Contemporary studies consistently identify digital transformation as a key determinant of:

- educational quality,

- operational efficiency,
- organizational agility,
- student engagement,
- institutional competitiveness.

However, empirical investigations focusing on public universities in Latin America remain scarce. Existing evidence rarely integrates educational quality and institutional efficiency within a unified analytical framework. This research addresses this limitation by examining both outcomes simultaneously through an institutional case study.

6. Quasi-Experimental Design

The research adopts a pragmatic mixed-methods case study supported by a quasi-experimental explanatory design.

Academic and administrative units with different levels of digital transformation will be compared using institutional indicators and perception-based measures. Quantitative data will identify statistically significant differences, whereas qualitative evidence will explain the organizational mechanisms responsible for those differences.

This design enables causal inference within naturally occurring institutional settings while maintaining ecological validity.

7. Data Management Plan Outline

The research will implement a comprehensive data management strategy based on the FAIR principles.

Primary data will include surveys, interviews, and focus groups. Secondary data will comprise institutional reports, digital transformation strategies, quality assurance documents, and performance indicators.

Data will be anonymized, encrypted, stored securely, and documented through standardized metadata procedures. Only authorized researchers will have access to identifiable information.

8. Ethical Considerations

The study will comply with international research ethics standards.

Participation will be voluntary and based on informed consent. Confidentiality, anonymity, and data privacy will be guaranteed throughout the project. Institutional authorization and ethics committee approval will be obtained before data collection.

Particular attention will be paid to minimizing organizational risks associated with institutional performance evaluation and protecting participants from potential professional repercussions.

9. Project Schedule / Timeline

The project is planned over twelve months.

- Months 1–2: Literature review and theoretical framework.
- Month 3: Research design and instrument development.
- Month 4: Pilot testing and instrument validation.
- Months 5–6: Quantitative and qualitative data collection.
- Months 7–8: Statistical and qualitative data analysis.
- Month 9: Integration and triangulation of findings.
- Months 10–11: Thesis writing and revision.
- Month 12: Final review, submission, and defense preparation.

10. Expected Results and Impact

The study is expected to demonstrate how digital transformation contributes to improving educational quality and institutional efficiency within a leading public university.

Scientifically, the research will provide an integrated framework linking digital transformation, educational quality, and institutional performance. Methodologically, it will validate a mixed-methods approach suitable for studying complex organizational phenomena in higher education.

Practically, the findings will support evidence-based decision-making for university leaders and policymakers, offering recommendations to strengthen digital governance, optimize institutional processes, and improve educational outcomes.

Finally, the proposed framework is expected to be transferable to other public universities in Peru and Latin America, contributing to sustainable digital transformation and institutional modernization in emerging higher education systems.